Direct Preference Optimization: Aligning Generative AI with Human Preferences, from Text to Images

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Overview

The Alignment Problem

The Alignment Problem

Large Language Models (LLMs)

Capability: Generate human-like text, code, and dialogue.
Failure: Can be unhelpful, factually incorrect, or produce biased/harmful content without alignment.

Diffusion Models

Capability: Generate photorealistic and artistic images from text prompts.
Failure: Often misinterpret complex prompts, produce aesthetic flaws, or generate not-safe-for-work (NSFW) imagery.

The Alignment Problem

The Core Question

Given a prompt x and two outputs y_w (winner) and y_l (loser) such that a human prefers y_w over y_l, how can we efficiently fine-tune a model to reflect this preference?

Reinforcement Learning from Human Feedback

Reinforcement Learning from Human Feedback

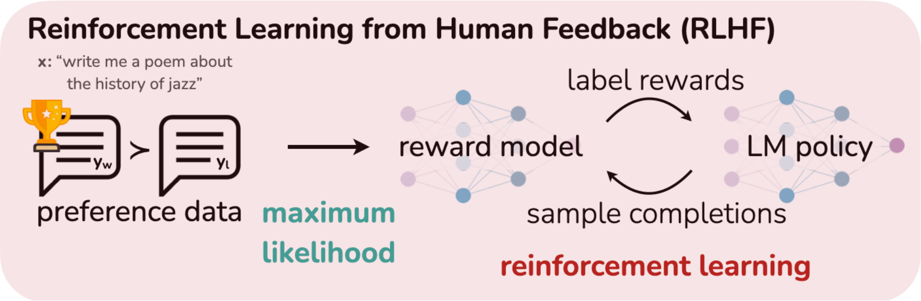

Figure: The standard three-stage RLHF pipeline.

Reinforcement Learning from Human Feedback

Reward Model (RM) Training

Bradley-Terry model:

$$\mathcal{D} = \left\{x^{(i)}, y_w^{(i)}, y_l^{(i)}\right\}_{i=1}^{N}$$

$$p^*(y_w \succ y_l \mid x) = \sigma\left(r^*(x, y_w) - r^*(x, y_l)\right)$$

Binary cross-entropy loss:

$$\mathcal{L}_R(r_\phi, \mathcal{D}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}}\left[\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))\right]$$

Reinforcement Learning from Human Feedback

RL Fine-Tuning

$$\max_{\pi_\theta} \mathbb{E}_{x\sim\mathcal{D}, y\sim\pi_\theta(y|x)}\left[r_\phi(x, y)\right] - \beta\mathbb{D}_{\mathrm{KL}}\left[\pi_\theta(y \mid x) \mid\mid \pi_{ref}(y \mid x)\right]$$

Reinforcement Learning from Human Feedback

Key Disadvantages

- **Training Instability**
- **Complex Pipeline**
- **Resource Intensive**

Direct Preference Optimization

Direct Preference Optimization

The optimal policy for the KL-constrained RL objective is:

$$\pi_r(y|x) = \frac{1}{Z(x)}\pi_{\text{ref}}(y|x)\exp\left(\frac{1}{\beta}r(x,y)\right)$$

$$Z(x) = \sum_{y} \pi_{ref}(y \mid x)\exp\left(\frac{1}{\beta}r(x,y)\right)$$

Direct Preference Optimization

We can rearrange this to define the reward function in terms of policies:

$$r(x, y) = \beta \log \frac{\pi_r(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x)$$

Direct Preference Optimization

Substitute this into the preference loss, which only depends on the reward *difference*:

$$r(x, y_w) - r(x, y_l) = \beta \log \frac{\pi_r(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_r(y_l|x)}{\pi_{\text{ref}}(y_l|x)}$$

Direct Preference Optimization

The "Magic" Step

The intractable partition function $Z(x)$ completely cancels out! This allows us to optimize the policy π_θ directly on preference data without ever fitting an explicit reward model.

The DPO Objective

This insight leads to a simple binary cross-entropy loss:

Direct Preference Optimization (DPO) Loss

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\hat{r}_\theta(x, y_w) - \hat{r}_\theta(x, y_l) \right) \right]$$

where the implicit reward $\hat{r}_\theta(x, y)$ is defined as:

$$\hat{r}_\theta(x, y) = \beta \log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

The DPO Objective

How it Works

The loss increases the relative log-probability of the preferred response (y_w) over the dispreferred response (y_l).

$$\nabla_\theta \mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{ref}) = -\beta \mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\Bigg[\underbrace{\sigma(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \Bigg[\underbrace{\nabla_\theta \log \pi(y_w \mid x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_\theta \log \pi(y_l \mid x)}_{\text{decrease likelihood of } y_l}\Bigg]\Bigg]$$

The DPO Objective

Advantages

- **Simple**
- **Stable**
- **Efficient**

DPO Performance on LLMs

DPO Performance on LLMs

Key Finding

DPO matches or exceeds the performance of PPO-based RLHF while being substantially simpler and more stable to train.

DPO Performance on LLMs

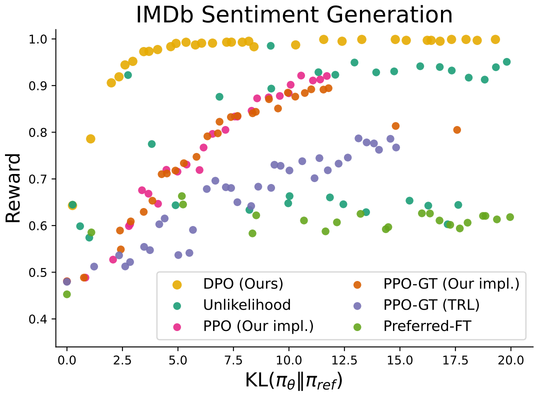

Figure: DPO provides the highest expected reward for all KL values

DPO Performance on LLMs

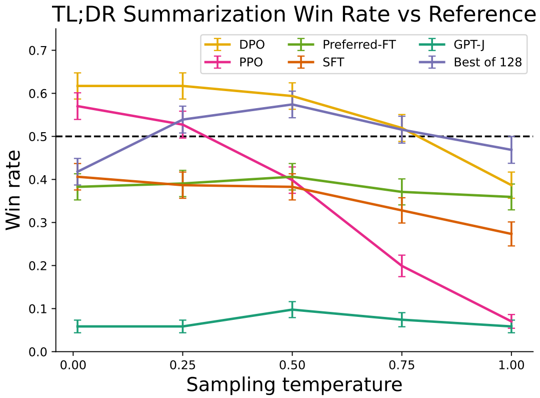

Figure: Win-rate against SFT model, evaluated by GPT-4.

DPO on Diffusion

Denoising Diffusion Models

Core Concept

Diffusion models learn to generate data by reversing a gradual noising process.

- **Forward Process:** An image x_0 is gradually destroyed by adding Gaussian noise over T timesteps, producing a noisy latent x_t.

$$q(x_t|x_0) = \mathcal{N}(x_t; \alpha_t x_0, \sigma_t^2 I)$$

- **Reverse Process:** A model ϵ_θ is trained to predict the noise that was added to create x_t. This defines a Markov chain that generates a clean image x_0 from pure noise x_T. Each step is a Gaussian transition:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

Denoising Diffusion Models

Training Objective

The model is trained to minimize the difference between the actual noise ϵ and the predicted noise $\epsilon_\theta(x_t, t)$ at each timestep. This is derived from the Evidence Lower Bound (ELBO).

$$L_{\text{DM}} = \mathbb{E}_{x_0,\epsilon,t,x_t}\left[\omega(\lambda_t)\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2\right]$$

RLHF on Diffusion

Reward Model

- **Reward Model:**

$$p_{\mathrm{BT}}(x_0^w \succ x_0^l | c) = \sigma(r(c, x_0^w) - r(c, x_0^l))$$

- **Binary cross-entropy loss:**

$$L_{\mathrm{BT}}(\phi) = -\mathbb{E}_{c, x_0^w, x_0^l}\left[\log \sigma\left(r_\phi(c, x_0^w) - r_\phi(c, x_0^l)\right)\right]$$

The RLHF Objective (Restated)

$$\max_{p_\theta} \mathbb{E}_{c \sim \mathcal{D}, x_0 \sim p_\theta(x_0|c)} [r(c, x_0)] - \beta \mathbb{D}_{\mathrm{KL}} [p_\theta(x_0|c) \| p_{\mathrm{ref}}(x_0|c)]$$

Restating the DPO Objective

The optimal policy p_θ^* for the RLHF objective:

$$p_\theta^*(x_0|c) = \frac{1}{Z(c)}(x_0|c)\exp\left(\frac{1}{\beta}r(c, x_0)\right)$$

DPO Loss (for models with tractable likelihoods)

$$L_{\text{DPO}}(\theta) = -\mathbb{E}_{(c,x_0^w,x_0^l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{p_\theta(x_0^w|c)}{(x_0^w|c)} - \beta\log\frac{p_\theta(x_0^l|c)}{(x_0^l|c)}\right)\right]$$

Adapting DPO to Diffusion Models: The Challenge

The Core Problem

This formulation directly depends on evaluating the likelihoods $p_\theta(x_0|c)$ and $(x_0|c)$, which is exactly what we cannot do efficiently for diffusion models.

The Solution

- Re-cast the DPO objective using the Evidence Lower Bound (ELBO) on the log-likelihood.
- This connects the intractable likelihood to the model's tractable training objective: the **denoising error**.

The Diffusion-DPO Objective

Core Idea

Instead of maximizing the likelihood ratio, we minimize the denoising error for the winning image relative to the losing image.

$$\text{Goal: } \text{Error(winner)} < \text{Error(loser)}$$

Diffusion-DPO Loss (Conceptual)

$$L_{\text{DPO-Diffusion}} \approx -\mathbb{E}\left[\log \sigma \left(\beta \left(\text{MSE}(y_l) - \text{MSE}(y_w)\right)\right)\right]$$

where $\text{MSE}(y) = \|\epsilon - \epsilon_\theta(y_t, t)\|_2^2$ is the denoising mean squared error for a noisy version y_t of image y.

Deriving the Diffusion-DPO Objective

1. Reframe the RL Objective

$$\max_{p_\theta} \mathbb{E}_{x_{0:T} \sim p_\theta(x_{0:T}|c)} \left[R(c, x_{0:T})\right] - \beta \mathbb{D}_{\text{KL}}[p_\theta ||]$$

2. Approximate the Path Sampling

$$p_\theta(x_{1:T}|x_0) \approx q(x_{1:T}|x_0)$$

3. Simplify to a Denoising Error Objective

$$L(\theta) = -\mathbb{E}_{(x_0^w, x_0^l) \sim \mathcal{D}, t \sim \mathcal{U}, \epsilon^w, \epsilon^l \sim \mathcal{N}} \log \sigma \left(-\beta' \left(\underbrace{\left(\|\epsilon^w - \epsilon_\theta(x_t^w, t)\|_2^2 - \|\epsilon^w - \epsilon_{\text{ref}}(x_t^w, t)\|_2^2 \right)}_{\text{Implicit reward for } x_w} - \underbrace{\left(\|\epsilon^l - \epsilon_\theta(x_t^l, t)\|_2^2 - \|\epsilon^l - \epsilon_{\text{ref}}(x_t^l, t)\|_2^2 \right)}_{\text{Implicit reward for } x_l} \right) \right)$$

The loss directly minimizes the relative denoising error for the preferred image.

The Diffusion-DPO Objective

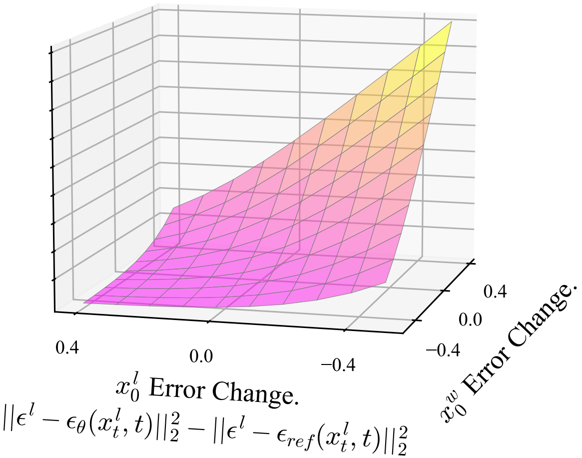

Figure: Loss Surface

Diffusion-DPO Performance

Diffusion-DPO Performance

Key Finding

Fine-tuning SDXL with Diffusion-DPO significantly improves visual appeal and prompt alignment.

Diffusion-DPO Output

Figure: Diffusion-DPO generates more visually appealing images.

Diffusion-DPO Performance

Figure: Human preference win rates of DPO-SDXL vs. baselines.

- The DPO-tuned base model (3.5B params) outperforms the much larger SDXL 1.0 + Refiner (6.6B params).

Qualitative Results: DPO-SDXL vs. SDXL

- **Better Prompt Alignment:** Follows complex instructions more accurately.
- **Enhanced Realism & Aesthetics:** Produces more photorealistic details and compelling artistic styles.
- **More Coherent Compositions:** Better object placement and interaction.

Related and Future Work

Related and Future Work

Related Work

- Builds upon a rich history of learning from preferences (e.g., Bradley-Terry models).
- Leverages the success of large-scale pre-training for generative models.
- Directly simplifies the now-standard RLHF alignment paradigm.

Related and Future Work

Future Directions

- **Personalization:** Fine-tuning models to individual artistic styles or writing preferences.
- **AI Feedback (RLAIF):** Using powerful models to generate preference labels at scale.
- **New Modalities:** Applying the DPO principle to video, 3D models, and music generation.

Conclusion

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Main Takeaway

Direct Preference Optimization (DPO) is a simple, stable, and highly effective paradigm for aligning generative models with human preferences.

- It elegantly **replaces the complex and unstable RLHF pipeline** with a direct classification loss.
- It was first shown to be highly effective for **language models**, matching or outperforming traditional RLHF (Rafailov et al., 2024).
- The core principle was successfully generalized to **diffusion models**, overcoming the challenge of intractable likelihoods to achieve state-of-the-art image quality and prompt alignment (Wallace et al., 2023).

References

[1] Rafailov, R., Sharma, A., Mitchell, E., Ermon, S., Manning, C. D., & Finn, C. (2024).
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[2] Wallace, B., Dang, M., Rafailov, R., Zhou, L., Lou, A., Purushwalkam, S., Ermon, S., Xiong, C., Joty, S., & Naik, N. (2023).
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The End